



5014 - Anatomy of a Low-Metallicity Starburst

Cycle: 3, Proposal Category: GO

INVESTIGATORS

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OBSERVATIONS

<i>Folder</i>	<i>Observation</i>	<i>Label</i>	<i>Observing Template</i>	<i>Science Target</i>
NGC5253				
	4	NGC5253 NIRCAM S W+LW combined	NIRCam Imaging	(1) NGC-5253
	19	NGC5253 NIRCAM S W+LW combined DUPLICATE with observed removed	NIRCam Imaging	(1) NGC-5253
	7	NGC5253 MIRI MRS	MIRI Medium Resolution Spectroscopy	(2) NGC-5253-MRS
	18	NGC5253 #12 MIRI MRS	MIRI Medium Resolution Spectroscopy	(5) NGC5253-12
	11	NGC5253 MIRI MRS Background	MIRI Medium Resolution Spectroscopy	(3) NGC-5253-BACKGROUND
	21	NGC5253 MIRI FULL ARRAY BACKGROUND	MIRI Imaging	(8) MIRI-sub-array-background
	8	NGC5253 MIRI FULL ARRAY	MIRI Imaging	(9) NGC-5253-copy
	12	NGC5253 MIRI SUB64	MIRI Imaging	(10) NGC5253-5-copy
	20	NGC5253 MIRI SUB64 BACKGROUND	MIRI Imaging	(8) MIRI-sub-array-background
	14	NGC5253 MRS-pointing NIRSpec/IFU Leakcal	NIRSpec IFU Spectroscopy	(7) NGC-5253-MRS-NoBck
	16	NGC5253 #5 NIRSpec/IFU Leakcal	NIRSpec IFU Spectroscopy	(4) NGC5253-5
	17	NGC5253 #12 NIRSpec/IFU Leakcal	NIRSpec IFU Spectroscopy	(6) NGC5253-12-NoBck

ABSTRACT

Dwarf galaxies, systems with low baryonic masses, are the most abundant population at any redshift. Dwarf systems likely played a major role during the Epoch of Reionization (EoR) because their shallow potential wells make them vulnerable to stellar feedback processes that can enhance the porosity of the interstellar medium (ISM). Reionization and the escape fraction of ionizing (Lyman Continuum, LyC) photons ultimately depend

on the ISM porosity. While fesc has been constrained on galactic scales in some high-redshift dwarf galaxies, it cannot be mapped at the critical few parsec scales around the young massive star clusters that drive feedback. Leveraging the parallels between high- z dwarfs and star-forming dwarf galaxies (SFDs) in the Local Universe, we propose a pilot study of the nearby iconic metal-poor SFD, NGC 5253, using NIRC*am* imaging, NIRS*pec*/IFU spectroscopy, MIRI imaging, and MRS/IFU spectroscopy. JWST will provide a detailed assessment of feedback on the surrounding ISM, with 2-7 pc resolution, through maps of ionized and warm molecular gas, Polycyclic Aromatic Hydrocarbons, and warm dust. JWST will also quantify dust production in the evolved stellar population, through a census of Asymptotic Giant Branch stars and the past star-formation history. These combined observations will yield a resolved measurement of fesc. This ambitious pilot program exploits JWST's superb infrared sensitivity and spatial resolution for a detailed spatially-resolved analysis that will revolutionize our understanding of stellar feedback, ISM porosity, and stellar dust production in a low-metallicity starburst environment, with potential similarities to conditions at the EoR.

OBSERVING DESCRIPTION

We will observe NGC5253 with NIRC*am* SW+LW imaging, NIRS*pec*/IFU, and MIRI imaging and MIRI/MRS IFU.

NIRS*pec*/IFU with G235H/F170LP and G395H/F270LP will be pointed at three groups of selected young, massive star clusters in the galaxy, and MIRI MRS/IFU spectral maps will be obtained for two pointings spanning 5 star clusters. We cover the entire MRS spectral range, using all 3 grating settings (SHORT-A, SHORT-B, SHORT-C). Dedicated background observations are planned, and simultaneous MIRI off-source imaging will be obtained for both MRS pointings (and the background MRS positions) in the same filters.

We will obtain deep NIRC*am* images with three filter pairs, F115W/F335M + F150W/F360M + F140M/F300M, as well as with two additional filter pairs, F187N/F405N + F182M/F430M. A 3-position MIRI mosaic will be acquired in four filters, F770W, F1000W, F1130W, and F2100W, and we also will obtain a SUB64 image with the same filter set as above. SUB64 avoids saturation for the brightest star cluster in all but a few pixels at the end of the ramp for F2100W.

We have fixed the PA for NIRC*am* and MIRI/MRS observations in order to ensure that the outer regions of the galaxy is well observed, and to acquire additional MIRI imaging on source during the MRS background observation.

Proposal 5014 - Targets - Anatomy of a Low-Metallicity Starburst

Fixed Targets	#	Name	Target Coordinates	Targ. Coord. Corrections	Miscellaneous
	(1)	NGC-5253	RA: 13 39 55.9632 (204.9831800d) Dec: -31 38 24.40 (-31.64011d) Equinox: J2000 <i>Comments: This object was generated by the targetselector and retrieved from the NED database.</i> <i>Category=Galaxy</i> <i>Description=[Dwarf irregular galaxies]</i> <i>Extended=YES</i>		
	(2)	NGC-5253-MRS	RA: 13 39 55.9195 (204.9829979d) Dec: -31 38 27.11 (-31.64086d) Equinox: J2000 <i>Comments:</i> <i>Category=Galaxy</i> <i>Description=[Dwarf irregular galaxies]</i> <i>Extended=YES</i>		
	(3)	NGC-5253-BACKGROUND	RA: 13 40 12.1362 (205.0505675d) Dec: -31 37 53.01 (-31.63139d) Equinox: J2000 <i>Comments:</i> <i>Category=Calibration</i> <i>Description=[Telescope/sky background]</i> <i>Extended=YES</i>		
	(4)	NGC5253-5	RA: 13 39 55.9938 (204.9833075d) Dec: -31 38 24.40 (-31.64011d) Equinox: J2000 <i>Comments:</i> <i>Category=Stellar Cluster</i> <i>Description=[Young star clusters]</i>		
	(5)	NGC5253-12	RA: 13 39 56.1500 (204.9839583d) Dec: -31 38 22.80 (-31.63967d) Equinox: J2000 <i>Comments:</i> <i>Category=Stellar Cluster</i> <i>Description=[Young star clusters]</i>		
	(6)	NGC5253-12-NoBck	RA: 13 39 56.1500 (204.9839583d) Dec: -31 38 22.80 (-31.63967d) Equinox: J2000 <i>Comments:</i> <i>Category=Stellar Cluster</i> <i>Description=[Young star clusters]</i>		
	(7)	NGC-5253-MRS-NoBck	RA: 13 39 55.9195 (204.9829979d) Dec: -31 38 27.11 (-31.64086d) Equinox: J2000 <i>Comments:</i> <i>Category=Galaxy</i> <i>Description=[Dwarf irregular galaxies]</i> <i>Extended=YES</i>		

Proposal 5014 - Targets - Anatomy of a Low-Metallicity Starburst

(8)	MIRI-sub-array-background	RA: 13 39 50.1400 (204.9589167d) Dec: -31 35 30.00 (-31.59167d) Equinox: J2000
Comments: Category=Unidentified Description=[Blank field]		
(9)	NGC-5253-copy	RA: 13 39 55.9632 (204.9831800d) Dec: -31 38 24.40 (-31.64011d) Equinox: J2000
Comments: This object was generated by the targetselector and retrieved from the NED database. Category=Galaxy Description=[Dwarf irregular galaxies] Extended=YES		
(10)	NGC5253-5-copy	RA: 13 39 55.9938 (204.9833075d) Dec: -31 38 24.40 (-31.64011d) Equinox: J2000
Comments: Category=Stellar Cluster Description=[Young star clusters]		

Proposal 5014 - Observation 4 - Anatomy of a Low-Metallicity Starburst

Observation	Proposal 5014, Observation 4: NGC5253 NIRCAM SW+LW combined									Mon Feb 03 21:00:12 GMT 2025	
	Diagnostic Status: Warning										
	Observing Template: NIRCam Imaging										
Diagnostics	(Visit 4:1) Warning (Form): Data Excess over lower threshold										
	(Visit 4:1) Warning (Form): Overheads are provisional until the Visit Planner has been run.										
Fixed Targets	#	Name	Target Coordinates			Targ. Coord. Corrections		Miscellaneous			
	(1)	NGC-5253	RA: 13 39 55.9632 (204.9831800d) Dec: -31 38 24.40 (-31.64011d) Equinox: J2000								
	Comments: This object was generated by the targetselector and retrieved from the NED database. Category=Galaxy Description=[Dwarf irregular galaxies] Extended=YES										
Template	Module		Subarray				Target Placement				
	ALL		FULL				Module Gap				
Dithers	#	Primary Dither Type		Primary Dithers		Subpixel Dither Type		Dither Size		Subpixel Positions	
	1	FULLBOX		4TIGHT		SMALL-GRID-DITHER				3	
Spectral Elements	#	Short Filter	Long Filter	Readout Pattern	Groups/Int	Integrations/Exp	Total Integrations	Total Dithers	Total Exposure Time	ETC Wkbk.Calc ID	
	1	F187N	F405N+F444W	SHALLOW4	5	1	12	12	3092.19	177994	
	2	F140M	F300M	SHALLOW4	5	1	12	12	3092.19	177990	
	3	F182M	F430M	SHALLOW4	5	1	12	12	3092.19	177994	
	4	F115W	F335M	SHALLOW4	5	1	12	12	3092.19	177990	
	5	F150W	F360M	SHALLOW4	5	1	12	12	3092.19	177990	
Special Requirements	Aperture PA Range 112 to 118 Degrees (V3 112.07457694 to 118.07457694)										

Proposal 5014 - Observation 19 - Anatomy of a Low-Metallicity Starburst

Observation	Proposal 5014, Observation 19: NGC5253 NIRCAM SW+LW combined DUPLICATE with observed removed									Mon Feb 03 21:00:12 GMT 2025
	Diagnostic Status: Warning									
	Observing Template: NIRCam Imaging									
Diagnostics	(Visit 19:1) Warning (Form): Overheads are provisional until the Visit Planner has been run.									
Fixed Targets	#	Name	Target Coordinates			Targ. Coord. Corrections			Miscellaneous	
	(1)	NGC-5253	RA: 13 39 55.9632 (204.9831800d) Dec: -31 38 24.40 (-31.64011d) Equinox: J2000							
	Comments: This object was generated by the targetselector and retrieved from the NED database. Category=Galaxy Description=[Dwarf irregular galaxies] Extended=YES									
Template	Module		Subarray				Target Placement			
	ALL		FULL				Module Gap			
Dithers	#	Primary Dither Type		Primary Dithers		Subpixel Dither Type		Dither Size		Subpixel Positions
	1	FULLBOX		4TIGHT		SMALL-GRID-DITHER				3
Spectral Elements	#	Short Filter	Long Filter	Readout Pattern	Groups/Int	Integrations/Exp	Total Integrations	Total Dithers	Total Exposure Time	ETC Wkbk.Calc ID
	1	F115W	F335M	SHALLOW4	5	1	12	12	3092.19	177990
	2	F150W	F360M	SHALLOW4	5	1	12	12	3092.19	177990
Special Requirements	Aperture PA Range 112 to 118 Degrees (V3 112.07457694 to 118.07457694)									

Proposal 5014 - Observation 7 - Anatomy of a Low-Metallicity Starburst

Observation	Proposal 5014, Observation 7: NGC5253 MIRI MRS												Mon Feb 03 21:00:12 GMT 2025
	Diagnostic Status: Warning												
	Observing Template: MIRI Medium Resolution Spectroscopy												
	Background Observations:[NGC5253 MIRI MRS Background (Obs 11)]												
Diagnostics	(NGC5253 MIRI MRS (Obs 7)) Warning (Form): Imager Filter overlap.												
	(Visit 7:1) Warning (Form): Overheads are provisional until the Visit Planner has been run.												
Fixed Targets	#	Name	Target Coordinates				Targ. Coord. Corrections				Miscellaneous		
	(2)	NGC-5253-MRS	RA: 13 39 55.9195 (204.9829979d) Dec: -31 38 27.11 (-31.64086d) Equinox: J2000										
	Comments: Category=Galaxy Description=[Dwarf irregular galaxies] Extended=YES												
Acquisition	#	Target											
	1	NONE											
Template	AcqFilter	Primary Channel				Simultaneous Imaging				Imager Subarray		Grating Wheel Direction	
	F1500W	All MRS				YES				FULL		Allow Auto Reorder	
Dithers	#	Dither Type				Optimized For				Direction			
	1	4-Point				EXTENDED SOURCE				NEGATIVE			
Spectral Elements	#	Wavelength Range	Detector	Filter	Readout Pattern	Groups/Int	Integrations/Exp	Exposures/Dith	Dither	Total Dithers	Total Integrations	Total Exposure Time	ETC Wkbk.Calc ID
	1		IMAGER	F2100W	FASTR1	10	4	1	Dither 1	4	16	477.307	177992
	1	SHORT(A)	MRSLONG		FASTR1	18	4	1	Dither 1	4	16	832.512	177992
	1	SHORT(A)	MRSSHORT		FASTR1	18	4	1	Dither 1	4	16	832.512	177992
	2		IMAGER	F1130W	FASTR1	10	4	1	Dither 1	4	16	477.307	177992
	2	MEDIUM(B)	MRSLONG		FASTR1	15	4	1	Dither 1	4	16	699.31	177992
	2	MEDIUM(B)	MRSSHORT		FASTR1	15	4	1	Dither 1	4	16	699.31	177992
	3		IMAGER	F770W	FASTR1	12	4	1	Dither 1	4	16	566.108	177992
	3	LONG(C)	MRSLONG		FASTR1	20	4	1	Dither 1	4	16	921.313	177992
	3	LONG(C)	MRSSHORT		FASTR1	20	4	1	Dither 1	4	16	921.313	177992

Proposal 5014 - Observation 7 - Anatomy of a Low-Metallicity Starburst

Special Requirements	Aperture PA Range 90 to 110 Degrees (V3 90.0 to 110.0) Sequence Observations 7, 11, 18, Non-interruptible
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Proposal 5014 - Observation 18 - Anatomy of a Low-Metallicity Starburst

Observation	Proposal 5014, Observation 18: NGC5253 #12 MIRI MRS												Mon Feb 03 21:00:12 GMT 2025	
	Diagnostic Status: Warning													
	Observing Template: MIRI Medium Resolution Spectroscopy													
	Background Observations:[NGC5253 MIRI MRS Background (Obs 11)]													
Diagnostics	(NGC5253 #12 MIRI MRS (Obs 18)) Warning (Form): Imager Filter overlap.													
	(Visit 18:1) Warning (Form): Overheads are provisional until the Visit Planner has been run.													
Fixed Targets	#	Name	Target Coordinates				Targ. Coord. Corrections				Miscellaneous			
	(5)	NGC5253-12	RA: 13 39 56.1500 (204.9839583d) Dec: -31 38 22.80 (-31.63967d) Equinox: J2000											
	Comments: Category=Stellar Cluster Description=[Young star clusters]													
Acquisition	#	Target												
	1	NONE												
Template	AcqFilter	Primary Channel				Simultaneous Imaging				Imager Subarray		Grating Wheel Direction		
	F1500W	All MRS				YES				FULL		Allow Auto Reorder		
Dithers	#	Dither Type				Optimized For				Direction				
	1	4-Point				EXTENDED SOURCE				NEGATIVE				
Spectral Elements	#	Wavelength Range	Detector	Filter	Readout Pattern	Groups/Int	Integrations/Exp	Exposures/Dith	Dither	Total Dithers	Total Integrations	Total Exposure Time	ETC Wkbk.Calc ID	
	1		IMAGER	F2100W	FASTR1	10	4	1	Dither 1	4	16	477.307	177992	
	1	SHORT(A)	MRSLONG		FASTR1	18	4	1	Dither 1	4	16	832.512	177992	
	1	SHORT(A)	MRSSHORT		FASTR1	18	4	1	Dither 1	4	16	832.512	177992	
	2		IMAGER	F1130W	FASTR1	10	4	1	Dither 1	4	16	477.307	177992	
	2	MEDIUM(B)	MRSLONG		FASTR1	15	4	1	Dither 1	4	16	699.31	177992	
	2	MEDIUM(B)	MRSSHORT		FASTR1	15	4	1	Dither 1	4	16	699.31	177992	
	3		IMAGER	F770W	FASTR1	12	4	1	Dither 1	4	16	566.108	177992	
	3	LONG(C)	MRSLONG		FASTR1	20	4	1	Dither 1	4	16	921.313	177992	
	3	LONG(C)	MRSSHORT		FASTR1	20	4	1	Dither 1	4	16	921.313	177992	

Proposal 5014 - Observation 18 - Anatomy of a Low-Metallicity Starburst

Special Requirements	Aperture PA Range 90 to 110 Degrees (V3 90.0 to 110.0) Sequence Observations 7, 11, 18, Non-interruptible
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Proposal 5014 - Observation 11 - Anatomy of a Low-Metallicity Starburst

Observation	Proposal 5014, Observation 11: NGC5253 MIRI MRS Background												Mon Feb 03 21:00:12 GMT 2025
	Diagnostic Status: Warning												
	Observing Template: MIRI Medium Resolution Spectroscopy												
	Background Observation For: [NGC5253 MIRI MRS (Obs 7), NGC5253 #12 MIRI MRS (Obs 18)]												
Diagnostics	(NGC5253 MIRI MRS Background (Obs 11)) Warning (Form): Imager Filter overlap.												
	(Visit 11:1) Warning (Form): Overheads are provisional until the Visit Planner has been run.												
Fixed Targets	#	Name	Target Coordinates				Targ. Coord. Corrections			Miscellaneous			
	(3)	NGC-5253-BACKGROUND	RA: 13 40 12.1362 (205.0505675d) Dec: -31 37 53.01 (-31.63139d) Equinox: J2000										
	Comments: Category=Calibration Description=[Telescope/sky background] Extended=YES												
Acquisition	#	Target											
	1	NONE											
Template	AcqFilter	Primary Channel				Simultaneous Imaging			Imager Subarray		Grating Wheel Direction		
	F1500W	All MRS				YES			FULL		Allow Auto Reorder		
Dithers	#	Dither Type				Optimized For				Direction			
	1	4-Point				BACKGROUND				NEGATIVE			
Spectral Elements	#	Wavelength Range	Detector	Filter	Readout Pattern	Groups/Int	Integrations/Exp	Exposures/Dith	Dither	Total Dithers	Total Integrations	Total Exposure Time	ETC Wkbk.Calc ID
	1		IMAGER	F2100W	FASTR1	10	4	1	Dither 1	4	16	477.307	177992
	1	SHORT(A)	MRSLONG		FASTR1	18	4	1	Dither 1	4	16	832.512	177992
	1	SHORT(A)	MRSSHORT		FASTR1	18	4	1	Dither 1	4	16	832.512	177992
	2		IMAGER	F1130W	FASTR1	10	4	1	Dither 1	4	16	477.307	177992
	2	MEDIUM(B)	MRSLONG		FASTR1	15	4	1	Dither 1	4	16	699.31	177992
	2	MEDIUM(B)	MRSSHORT		FASTR1	15	4	1	Dither 1	4	16	699.31	177992
	3		IMAGER	F770W	FASTR1	12	4	1	Dither 1	4	16	566.108	177992
	3	LONG(C)	MRSLONG		FASTR1	20	4	1	Dither 1	4	16	921.313	177992
	3	LONG(C)	MRSSHORT		FASTR1	20	4	1	Dither 1	4	16	921.313	177992

Proposal 5014 - Observation 11 - Anatomy of a Low-Metallicity Starburst

Special Requirements	Aperture PA Range 90 to 110 Degrees (V3 90.0 to 110.0) Sequence Observations 7, 11, 18, Non-interruptible
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Proposal 5014 - Observation 21 - Anatomy of a Low-Metallicity Starburst

Observation	Proposal 5014, Observation 21: NGC5253 MIRI FULL ARRAY BACKGROUND									Mon Feb 03 21:00:12 GMT 2025	
	Diagnostic Status: Warning										
	Observing Template: MIRI Imaging										
	Background Observation For: [NGC5253 MIRI FULL ARRAY (Obs 8)]										
Diagnostics	(Visit 21:1) Warning (Form): Overheads are provisional until the Visit Planner has been run.										
	(Visit 21:1) Informational (Form): Visit schedulable, but most scheduling windows are when JWST is pointed in direction of greatest micrometeoroid impact risk. This is likely due to scheduling special requirements.										
Fixed Targets	#	Name	Target Coordinates			Targ. Coord. Corrections		Miscellaneous			
	(8)	MIRI-sub-array-background	RA: 13 39 50.1400 (204.9589167d) Dec: -31 35 30.00 (-31.59167d) Equinox: J2000								
	Comments: Category=Unidentified Description=[Blank field]										
Template	Subarray										
	FULL										
Dithers	#	Dither Type	Starting Point	Number of Points	Points	Starting Set	Number of Sets	Optimized For	Direction	Pattern Size	
	1	CYCLING	1	4	4	5	1			LARGE	
Spectral Elements	#	Filter	Readout Pattern	Groups/Int	Integrations/Exp	Exposures/Dith	Dither	Total Dithers	Total Integrations	Total Exposure Time	ETC Wkbk.Calc ID
	1	F770W	FASTR1	12	4	1	Dither 1	4	16	566.108	177992
	2	F1000W	FASTR1	10	4	1	Dither 1	4	16	477.307	177992
	3	F1130W	FASTR1	10	4	1	Dither 1	4	16	477.307	177992
	4	F2100W	FASTR1	10	4	1	Dither 1	4	16	477.307	177992
Special Requirements	Aperture PA Range 286 to 319 Degrees (V3 281.16455103 to 314.16455103)										
	Sequence Observations 8, 21, Non-interruptible										

Proposal 5014 - Observation 8 - Anatomy of a Low-Metallicity Starburst

Observation	Proposal 5014, Observation 8: NGC5253 MIRI FULL ARRAY										Mon Feb 03 21:00:12 GMT 2025
	Diagnostic Status: Warning										
	Observing Template: MIRI Imaging										
	Background Observations:[NGC5253 MIRI FULL ARRAY BACKGROUND (Obs 21)]										
Diagnostics	(NGC5253 MIRI FULL ARRAY (Obs 8)) Warning (Form): Target requiring background exposure selected for template that doesn't require background exposure										
	(Visit 8:1) Warning (Form): Overheads are provisional until the Visit Planner has been run.										
	(Visit 8:2) Warning (Form): Overheads are provisional until the Visit Planner has been run.										
	(Visit 8:3) Warning (Form): Overheads are provisional until the Visit Planner has been run.										
	(Visit 8:1) Informational (Form): Visit schedulable, but most scheduling windows are when JWST is pointed in direction of greatest micrometeoroid impact risk. This is likely due to scheduling special requirements.										
	(Visit 8:2) Informational (Form): Visit schedulable, but most scheduling windows are when JWST is pointed in direction of greatest micrometeoroid impact risk. This is likely due to scheduling special requirements.										
	(Visit 8:3) Informational (Form): Visit schedulable, but most scheduling windows are when JWST is pointed in direction of greatest micrometeoroid impact risk. This is likely due to scheduling special requirements.										
Fixed Targets	#	Name	Target Coordinates			Targ. Coord. Corrections			Miscellaneous		
	(9)	NGC-5253-copy	RA: 13 39 55.9632 (204.9831800d) Dec: -31 38 24.40 (-31.64011d) Equinox: J2000								
	Comments: This object was generated by the targetselector and retrieved from the NED database.										
	Category=Galaxy										
	Description=[Dwarf irregular galaxies] Extended=YES										
Template	Subarray										
	FULL										
Mosaic	Rows	Columns	Row Overlap %		Column Overlap %		Row shift (deg)		Column shift (deg)		Tile Order
	1	3	15.789		15.789		0.0		0.0		DEFAULT
Dithers	#	Dither Type	Starting Point	Number of Points	Points	Starting Set	Number of Sets	Optimized For	Direction	Pattern Size	
	1	CYCLING	1	4	4	5	1			LARGE	
Spectral Elements	#	Filter	Readout Pattern	Groups/Int	Integrations/Exp	Exposures/Dith	Dither	Total Dithers	Total Integrations	Total Exposure Time	ETC Wkbk.Calc ID
	1	F770W	FASTR1	12	4	1	Dither 1	4	16	566.108	177992
	2	F1000W	FASTR1	10	4	1	Dither 1	4	16	477.307	177992
	3	F1130W	FASTR1	10	4	1	Dither 1	4	16	477.307	177992
	4	F2100W	FASTR1	10	4	1	Dither 1	4	16	477.307	177992

Proposal 5014 - Observation 8 - Anatomy of a Low-Metallicity Starburst

Special Requirements	Group Visits within 53.0 Days Aperture PA Range 286 to 319 Degrees (V3 281.16455103 to 314.16455103) Visits Same PA Sequence Observations 8, 21, Non-interruptible
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Proposal 5014 - Observation 12 - Anatomy of a Low-Metallicity Starburst

Observation	Proposal 5014, Observation 12: NGC5253 MIRI SUB64										Mon Feb 03 21:00:12 GMT 2025
	Diagnostic Status: Warning										
	Observing Template: MIRI Imaging										
	Background Observations:[NGC5253 MIRI SUB64 BACKGROUND (Obs 20)]										
Diagnostics	(NGC5253 MIRI SUB64 (Obs 12)) Warning (Form): Target requiring background exposure selected for template that doesn't require background exposure										
	(Visit 12:1) Warning (Form): Overheads are provisional until the Visit Planner has been run.										
Fixed Targets	#	Name	Target Coordinates			Targ. Coord. Corrections			Miscellaneous		
	(10)	NGC5253-5-copy	RA: 13 39 55.9938 (204.9833075d) Dec: -31 38 24.40 (-31.64011d) Equinox: J2000								
	Comments: Category=Stellar Cluster Description=[Young star clusters]										
Template	Subarray										
	SUB64										
Dithers	#	Dither Type	Starting Point	Number of Points	Points	Starting Set	Number of Sets	Optimized For	Direction	Pattern Size	
	1	4-Point-Sets				1	1	POINT SOURCE	POSITIVE	DEFAULT	
Spectral Elements	#	Filter	Readout Pattern	Groups/Int	Integrations/Exp	Exposures/Dith	Dither	Total Dithers	Total Integrations	Total Exposure Time	ETC Wkbk.Calc ID
	1	F770W	FASTR1	5	1	1	Dither 1	4	4	1.702	177992
	2	F1000W	FASTR1	5	1	1	Dither 1	4	4	1.702	177992
	3	F1130W	FASTR1	5	1	1	Dither 1	4	4	1.702	177992
	4	F2100W	FASTR1	5	1	1	Dither 1	4	4	1.702	177992
Special Requirements	Sequence Observations 12, 20, Non-interruptible										

Proposal 5014 - Observation 20 - Anatomy of a Low-Metallicity Starburst

Observation	Proposal 5014, Observation 20: NGC5253 MIRI SUB64 BACKGROUND										Mon Feb 03 21:00:12 GMT 2025
	Diagnostic Status: Warning										
	Observing Template: MIRI Imaging										
	Background Observation For: [NGC5253 MIRI SUB64 (Obs 12)]										
Diagnostics	(Visit 20:1) Warning (Form): Overheads are provisional until the Visit Planner has been run.										
Fixed Targets	#	Name	Target Coordinates			Targ. Coord. Corrections			Miscellaneous		
	(8)	MIRI-sub-array-background	RA: 13 39 50.1400 (204.9589167d)								
			Dec: -31 35 30.00 (-31.59167d)								
			Equinox: J2000								
Template	Comments: Category=Unidentified Description=[Blank field]										
	Subarray SUB64										
Dithers	#	Dither Type	Starting Point	Number of Points	Points	Starting Set	Number of Sets	Optimized For	Direction	Pattern Size	
	1	4-Point-Sets				1	1	POINT SOURCE	POSITIVE	DEFAULT	
Spectral Elements	#	Filter	Readout Pattern	Groups/Int	Integrations/Exp	Exposures/Dith	Dither	Total Dithers	Total Integrations	Total Exposure Time	ETC Wkbk.Calc ID
	1	F770W	FASTR1	5	1	1	Dither 1	4	4	1.702	177992
	2	F1000W	FASTR1	5	1	1	Dither 1	4	4	1.702	177992
	3	F1130W	FASTR1	5	1	1	Dither 1	4	4	1.702	177992
	4	F2100W	FASTR1	5	1	1	Dither 1	4	4	1.702	177992
Special Requirements	Sequence Observations 12, 20, Non-interruptible										

Proposal 5014 - Observation 14 - Anatomy of a Low-Metallicity Starburst

Observation	Proposal 5014, Observation 14: NGC5253 MRS-pointing NIRSpec/IFU Leakcal											Mon Feb 03 21:00:12 GMT 2025
	Diagnostic Status: Warning Observing Template: NIRSpec IFU Spectroscopy											
Diagnostics	(Visit 14:1) Warning (Form): Overheads are provisional until the Visit Planner has been run.											
Fixed Targets	#	Name	Target Coordinates				Targ. Coord. Corrections			Miscellaneous		
	(7)	NGC-5253-MRS-NoBck	RA: 13 39 55.9195 (204.9829979d) Dec: -31 38 27.11 (-31.64086d) Equinox: J2000 Comments: Category=Galaxy Description=[Dwarf irregular galaxies] Extended=YES									
Template	TA Method						HFF Readout Mode					
	NONE						false					
Dithers	#	Dither Type		Size		Starting Point		Number of Points		Points		
	1	CYCLING		SMALL		1		6				
Spectral Elements	#	Grating/Filter	Readout Pattern	Groups/Int	Integrations/Exp	Leakcal	Dither	Autocal	Total Dithers	Total Integrations	Total Exposure Time	ETC Wkbk.Calc ID
	1	G235H/F170LP	NRSIRS2RAPID	10	1	false	true	NONE	6	6	962.867	177991
	2	G235H/F170LP	NRSIRS2RAPID	10	1	true	true	NONE	6	6	962.867	177991
	3	G395H/F290LP	NRSRAPID	4	1	false	true	NONE	6	6	322.103	177991
	4	G395H/F290LP	NRSRAPID	4	1	true	true	NONE	6	6	322.103	177991

Proposal 5014 - Observation 16 - Anatomy of a Low-Metallicity Starburst

Observation	Proposal 5014, Observation 16: NGC5253 #5 NIRSpec/IFU Leakcal											Mon Feb 03 21:00:12 GMT 2025
	Diagnostic Status: Warning											
Diagnostics	Observing Template: NIRSpec IFU Spectroscopy											
	(Visit 16:1) Warning (Form): Overheads are provisional until the Visit Planner has been run.											
Fixed Targets	#	Name	Target Coordinates				Targ. Coord. Corrections			Miscellaneous		
	(4)	NGC5253-5	RA: 13 39 55.9938 (204.9833075d) Dec: -31 38 24.40 (-31.64011d) Equinox: J2000									
Template	Comments: Category=Stellar Cluster Description=[Young star clusters]											
	TA Method						HFF Readout Mode					
Dithers	NONE						false					
	#	Dither Type			Size		Starting Point		Number of Points		Points	
Spectral Elements	1	CYCLING			SMALL		1		6			
	#	Grating/Filter	Readout Pattern	Groups/Int	Integrations/Exp	Leakcal	Dither	Autocal	Total Dithers	Total Integrations	Total Exposure Time	ETC Wkbk.Calc ID
	1	G235H/F170LP	NRSIRS2RAPID	10	1	false	true	NONE	6	6	962.867	177991
	2	G235H/F170LP	NRSIRS2RAPID	10	1	true	true	NONE	6	6	962.867	177991
	3	G395H/F290LP	NRSRAPID	4	1	false	true	NONE	6	6	322.103	177991
4	G395H/F290LP	NRSRAPID	4	1	true	true	NONE	6	6	322.103	177991	

Proposal 5014 - Observation 17 - Anatomy of a Low-Metallicity Starburst

Observation	Proposal 5014, Observation 17: NGC5253 #12 NIRSpec/IFU Leakcal											Mon Feb 03 21:00:12 GMT 2025
	Diagnostic Status: Warning											
Diagnostics	Observing Template: NIRSpec IFU Spectroscopy											
	(Visit 17:1) Warning (Form): Overheads are provisional until the Visit Planner has been run.											
Fixed Targets	#	Name	Target Coordinates				Targ. Coord. Corrections			Miscellaneous		
	(6)	NGC5253-12-NoBck	RA: 13 39 56.1500 (204.9839583d) Dec: -31 38 22.80 (-31.63967d) Equinox: J2000									
Template	Comments: Category=Stellar Cluster Description=[Young star clusters]											
	TA Method						HFF Readout Mode					
Dithers	NONE						false					
	#	Dither Type			Size		Starting Point		Number of Points		Points	
Spectral Elements	1	CYCLING			SMALL		1		6			
	#	Grating/Filter	Readout Pattern	Groups/Int	Integrations/Exp	Leakcal	Dither	Autocal	Total Dithers	Total Integrations	Total Exposure Time	ETC Wkbk.Calc ID
	1	G235H/F170LP	NRSIRS2RAPID	10	1	false	true	NONE	6	6	962.867	177991
	2	G235H/F170LP	NRSIRS2RAPID	10	1	true	true	NONE	6	6	962.867	177991
	3	G395H/F290LP	NRSRAPID	4	1	false	true	NONE	6	6	322.103	177991
4	G395H/F290LP	NRSRAPID	4	1	true	true	NONE	6	6	322.103	177991	